



Risky business: The impact of climate and climate variability on human population dynamics in Western Europe during the Last Glacial Maximum



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ARTICLE INFO

Article history:

Received 25 October 2016

Received in revised form

18 March 2017

Accepted 1 April 2017

Available online 14 April 2017

Keywords:

Pleistocene

Paleoclimate modelling

Western Europe

Random Forest

Upper Palaeolithic

Last Glacial Maximum

Population dynamics

ABSTRACT

The extent to which climate change has affected the course of human evolution is an enduring question. The ability to maintain spatially extensive social networks and a fluid social structure allows human foragers to “map onto” the landscape, mitigating the impact of ecological risk and conferring resilience. But what are the limits of resilience and to which environmental variables are foraging populations sensitive? We address this question by testing the impact of a suite of environmental variables, including climate variability, on the distribution of human populations in Western Europe during the Last Glacial Maximum (LGM). Climate variability affects the distribution of plant and animal resources unpredictably, creating an element of risk for foragers for whom mobility comes at a cost. We produce a model of habitat suitability that allows us to generate predictions about the probable distribution of human populations and discuss the implications of these predictions for the structure of human populations and their social and cultural evolution during the LGM.

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1. Introduction

The Last Glacial period was marked by pronounced climate instability and decreasing global temperatures, culminating with the Last Glacial Maximum (LGM) from 19,000 to 23,000 calibrated radiocarbon years Before Present (cal. B.P.) (Mix et al., 2001). Coinciding with Greenland Stadial 2.1 (Rasmussen et al., 2014), the LGM marks the maximum global ice volume and lowest sea stand during the last Glacial period (Mix et al., 2001). Climate conditions during this interval are thought to have significantly affected human population dynamics in Europe (Banks et al., 2013; Bocquet-Appel et al., 2005; Bocquet-Appel and Demars, 2000; d'Errico and Sánchez Goñi, 2003; Gamble et al., 2004; Müller et al., 2011; Tal-avaara et al., 2015; Tzedakis et al., 2007; van Andel and Davies,

2003; Verpoorte, 2009). Genetic evidence supports the existence of a Western Eurasian meta-population (including Western Europe) by 33,000 cal. B.P. (Fu et al., 2014, 2016; Posth et al.; Seguin-Orlando et al., 2014; Skorecki and Behar, 2013). Palaeogenetic data, however, indicate that large-scale population displacements occurred across Europe during the LGM (Fu et al., 2016) resulting in a loss of genetic diversity, followed by genetic turnover (Posth et al.; Soares et al., 2010). This reconstruction is consistent with paleontological evidence for the emergence of a modern European morphotype after 19,000 cal. B.P. (Churchill et al., 2000). The genetic evidence, together with the spatial distribution of archaeological sites and increasing regionalisation of Palaeolithic cultures (Gamble et al., 2004), suggest that the Western Eurasian meta-population became fragmented during the LGM. The role of climate change in shaping the social and biological history of modern human populations in Europe, therefore, is of considerable interest to scientists seeking to interpret the archaeological record.

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Archaeological evidence suggests that modern humans first entered Europe from Western Asia between 45,000 and 38,000 cal. B.P., travelling along two initial dispersal routes: west along the Danube valley to the Swabian Jura and from there south along the Rhône valley towards the Mediterranean coast; and west along the northern Mediterranean coastline (Conard et al., 2006; Conard, 2002; Conard and Bolus, 2003; Higham et al., 2012; Hublin, 2014; Mellars, 2006). This reconstruction of the timing and direction of human dispersals into Europe rests on the assumption that the early Aurignacian is the product of dispersing groups of modern humans (Davies, 2001, 2007), an assumption recently supported by physical and chronological evidence (Benazzi et al., 2015; Higham et al., 2011; Hublin, 2014). The geographical distribution of archaeological sites suggests that human populations in Europe retreated south of 49°N during the LGM (Verpoorte, 2009), contracting their range towards the southern Mediterranean peninsulas which acted as glacial refugia (Bailey et al., 2008; Gamble et al., 2004; Jennings et al., 2011; Jochim, 1987; Straus, 2015). At the same time, the archaeological record reveals the increased regionalisation of material culture (Gamble et al., 2004; Straus, 2000). Regional cultural differences existed previously, during the early Upper Palaeolithic (Vanhaeren and d'Errico, 2006), but with the onset of the LGM material culture production takes on a markedly regional expression, illustrated by the development of the Solutrean culture in Franco-Cantabria (Renard, 2011).

A full and convincing explanation for the successful initial expansion of modern human populations and the apparently rapid re-colonisation of northern Europe following the LGM remains elusive. It has been suggested that modern humans were dispersible and quick to take advantage of climatic upturns (Conard, 2011; Grove, 2015; Müller et al., 2011; Pinhasi et al., 2011; Tzedakis et al., 2007). This would have conferred an advantage during periods of climate instability, such as the period preceding the LGM and the period immediately following it, as the climate began to warm again. A reorganisation of human society with the “release from proximity” (Gamble, 1998), i.e., the ability to maintain relationships beyond the constraints of physical proximity, enabled the creation of spatially extensive social networks and may well have equipped modern human populations with a heightened ability to deal with environmental instability, or ecological risk, by facilitating information and resource sharing (Burke, 2012; Whallon et al., 2011). These hypotheses cannot be assessed, however, until we have identified a means of quantifying ecological risk under past climate regimes and fully tested the sensitivity of human systems to it, alone and in combination with topography and climate.

1.1. Hunter-gatherers and risk

Upper Palaeolithic populations pursued a mobile, hunting and gathering way of life. As (Kaplan, 2000) states on p.311: “One of the perennial problems confronted by virtually all hunter-gatherers is not only the seasonal variation in resources, but more significantly the periodic failure of all major resources”. The ethnographic, ethnohistorical and archaeological records suggest that hunter-gatherers are capable of adapting to a wide range of conditions. Seasonal variability may be a recurrent problem, but to the extent that it is predictable it doesn't constitute risk. Seasonal fluctuations in productivity can be anticipated and counter-measures applied, generating recurring patterns in the archaeological record (e.g., patterns of seasonal mobility and site occupation). Unpredictable variation in the distribution of resources, on the other hand, constitutes ecological risk since it means the outcome of foraging behaviour is uncertain (Winterhalder et al., 1999). Ecological risk, therefore, is not the same thing as resource scarcity. Unpredictable shifts in productivity are more difficult to manage and their impact

is scale dependant. For example, whereas the failure of a single resource may be countered by substituting other resources, the failure of a key resource (or multiple resource failure) may force a spatial reorganisation of the hunter-gatherer system.

The ability to deploy a variety of strategies to mitigate ecological risk is deeply embedded in the social, technological and economic structure of human hunting and gathering groups. Rather than simply engaging the world around them on a day-to-day basis, hunter-gatherers anticipate and actively evaluate strategies that minimize risk over the long term (Kelly, 2013) p.164. These strategies include technological innovation, such as the introduction of storage mechanisms to even out energy imbalances (Rowley-Conwy and Zvelebil, 1989), changes in diet (Stiner, 2001; Stutz et al., 2009; Winterhalder, 1981), the use of social mechanisms such as food-sharing and reciprocity (Bird-David, 1992; Cashdan, 1985; Smith, 1988; Weissner, 1982; Winterhalder, 1986; Winterhalder, 2001) and of course, mobility (Binford, 1981; Grove, 2009, 2010; Morgan, 2009; Yellen, 1986). These risk-reducing strategies allow human systems to manage resource fluctuations while minimising disruption - in other words, they confer resilience. Risk management strategies have social and emotional costs, however, and may affect the size and cohesiveness of hunter-gatherer social groups (Kelly, 2013) as well as other aspects of human culture (Collard et al., 2011). One expects, therefore, that trade-offs between different strategies will be carefully weighed (Winterhalder et al., 1999).

Repositioning people on the landscape is a common strategy for countering ecological risk (Morgan, 2009) and the nature of the response is linked to the scale of the risk (Collard et al., 2011). Hunter-gatherers may counter short-term variability in the distribution of resources by shifting from logistical to residential mobility strategies, for example (Morgan, 2009). In a residential strategy, base camps are abandoned when the resources within their local catchment are exhausted (*sensu* (Vita-Finzi and Higgs, 1970)), whereas a logistical strategy extends the effective catchment of base camps through the use of special purpose camps, located some distance away and exploited by small work parties (Binford, 1982). At a larger scale, however, ecological variability may induce more disruptive changes which come at a greater cost, both socially (as local groups disband) and physiologically, because mobility has energetic costs commensurate with the distances involved (Kelly, 1992) and the structure of the environment. In addition, mobility carries an element of risk, due to incomplete knowledge of conditions at distal locations (Winterhalder et al., 1999), that is scaled to distance: the further away the move, the greater the risk. Information sharing removes some, but not all of the uncertainty associated with residential mobility (Kelly, 1992; Whallon, 2006; Whallon et al., 2011). Ultimately, accurately predicting resource availability at distal locations is only possible if climate conditions generate consistent and predictable patterns.

The distribution of archaeological sites in Europe indicates human range contractions during the LGM, suggesting that environmental conditions exceeded the ability of hunter-gatherer systems to adapt and disrupted their spatial organisation. In this research, we aim to develop a robust and parsimonious predictive statistical model of habitat suitability which will allow us to identify the key environmental stressors that affected human population dynamics during the LGM and investigate the role of ecological risk in shaping their spatial distribution.

1.2. Climate change, climate variability and risk

Climate affects human systems at different spatial and temporal scales. Climate change, which refers to changes in climate conditions occurring on a decadal scale or greater, is thought to have

influenced long-term evolutionary processes affecting our lineage (Grove, 2012; Potts, 1998), triggering hominin dispersals and causing shifts in the geographic range of our species (Cupillard et al., 2013; Mellars, 1998; van Andel and Davies, 2003). Large swaths of Northern Europe became uninhabitable during the LGM as ice-sheets engulfed the landscape, for example, doubtless displacing human populations, while at the same time new land masses emerged as sea levels dropped. Abrupt climate events, such as the Heinrich events (which occur at a millennial scale) punctuate global glacial cycles and are thought to have disrupted hominin systems particularly during the late Pleistocene, when they may have contributed to the extinction of the Neanderthals (Banks et al., 2011, 2013; Bradtmöller et al., 2012; d'Errico and Sánchez Goni, 2003; Müller et al., 2011; Schmidt et al., 2012; Sepulchre et al., 2007; Tzedakis et al., 2007).

Climate variability, on the other hand, refers to small-scale, stochastic fluctuations around the mean climate trend, which may also have influenced the course of hominin evolution (Grove, 2011; Potts, 2013). Unpredictable patterns of resource availability generated by climate variability create social tensions in human populations that may resolve themselves peacefully, e.g., through group fission, or violently, through conflict (Ember and Ember, 1992). Climate variability correlates with periods of social strife in environmentally vulnerable regions today, for example (O'Loughlin et al., 2014). Stochastic properties of the environment condition the degree to which organisms become risk averse (Zhang et al., 2014) and we might expect, therefore, that the highly variable climate conditions observed during the late Pleistocene would foster the adoption of risk-averse strategies by human foragers (Morgan, 2009); this is a premise we test in this research.

Because climate change and climate variability are expected to play a role in determining the distribution of hominin populations, we include both types of predictor in the analysis that follows, which uses Random Forest and an iterative, automated selection approach to identify a suite of geographical and environmental predictors that affected the distribution of human populations living in Western Europe during the LGM.

2. Materials and methods

The spatial location of archaeological sites is used here as a proxy for the distribution of human populations, which we examine as a function of geographical and climatological predictors. Climate proxies, such as pollen counts, are a useful means of reconstructing past environments but reconciling different proxies can be difficult and uneven sampling distributions and poor chronological control make it difficult to work with large spatial extents and complicate the task of quantifying climate variability. For this reason, we use simulated climate variables, downscaled to a fine spatial resolution.

2.1. The archaeological sample

A sample of Western European archaeological sites was created by cross-referencing three existing databases: PACEA (d'Errico et al., 2011), S2Ages (Pettitt et al., 2003) and Climex (Margottini and Vai, 2004). We extracted archaeological occupations with ^{14}C dates falling between 19 and 23,000 cal years B.P., calibrated using OxCal 4.2, (Ramsey, 2009), from the existing databases. The resulting list was filtered using the following criteria: a) the radiocarbon date derives from an archaeological level and is consistent with the cultural attribution of the level; b) the validity of the date or the stratigraphic integrity of the archaeological level from which the sample was obtained is not discredited in subsequent publications; c) the proposed date is consistent with other

dates from the same stratigraphic sequence. Older dating protocols tend to produce radiocarbon dates with large margins of error, although it is now possible to obtain radiocarbon dates with much tighter margins of error (Davies et al., 2015; Higham et al., 2014; Hublin, 2014; Wood et al., 2013) we retained sites with conventional ^{14}C dates provided they have relatively tight margins of error and met the other selection criteria or where there is strong support, in the form of environmental or stratigraphic data, for an LGM attribution (see S1). We also added a number of recently published LGM sites meeting the selection criteria. The spatial coordinates of the sites were verified by contacting regional authorities (the DRAC for French sites), individual excavators or, as a last resort, visually checking coordinates in Google Earth against published descriptions of site locations. The resulting database contains 97 georeferenced archaeological sites dated to the LGM (Fig. 1; S1 Table).

2.2. Climate simulation

The complexity of the demographic patterns visible in the archaeological record suggests that local processes are at work, in addition to the broader spatial patterns produced by long-term climate trends (Burke et al., 2014). Previous research on human population dynamics during the late Pleistocene was limited in its ability to examine local scale patterning due to the spatial scale of the climate data available (Bradtmöller et al., 2012; Davies and Gollop, 2003; Gamble et al., 2004; Schmidt et al., 2012; Sepulchre et al., 2007; Tzedakis et al., 2007; van Andel and Davies, 2003). Recent advances in climate science allow us to explore climate change at much higher resolution (Schmatz et al., 2015) in addition to facilitating the incorporation of climate variability into our analyses (Brierley, 2015; Giorgi et al., 2004).

We begin with the Atmosphere-Ocean General Circulation Model (AOGCM) (IPSL-CM5A-LR) (Dufresne et al., 2013) with a spatial resolution of $\sim 1.9^\circ$ in latitude and 3.75° in longitude, with which we ran a global climate simulation for the LGM according to the PMIP3 protocol (Braconnot et al., 2012; Kageyama et al., 2013). We extracted a 50-year time series of monthly averages from the simulation output, including air temperature at 2 m above the surface, sea level pressure, surface wind, relative humidity and cloud cover fraction, and applied a statistical downscaling model to the data (GAM) (Vrac et al., 2007), obtaining high-resolution 2 m temperature and precipitation with a spatial scale of $\sim 15 \times 15$ km (Latombe et al. In prep). This downscaling procedure has been calibrated and applied to a European domain; we use the data produced for a domain defined by longitudes between -11.578000 dd and 24.73801 dd and latitudes between 32.720001 dd and 59.861000 dd. The spatial extent of the resulting climate data extends to the LGM coastal margins, but the rasters were clipped using a Holocene coastal mask because currently submerged Pleistocene land masses (e.g., Doggerland) have not been comprehensively surveyed and presence data (archaeological sites) are therefore lacking.

2.3. Predictive variables

Candidate predictors for archaeological site location include geographical variables, climate variables and climate variability indices. The geographical variables (elevation, slope, aspect) were derived from the SRTM 90-m digital elevation model sampled at a 1×1 km scale in arcGIS 10.3. Aspect was calculated and reclassified into five classes (one for each of the cardinal directions and a fifth class for flat surfaces). We use the downscaled climate outputs to calculate minima, maxima and average values for temperature and precipitation on monthly, seasonal and annual scales. Both minima

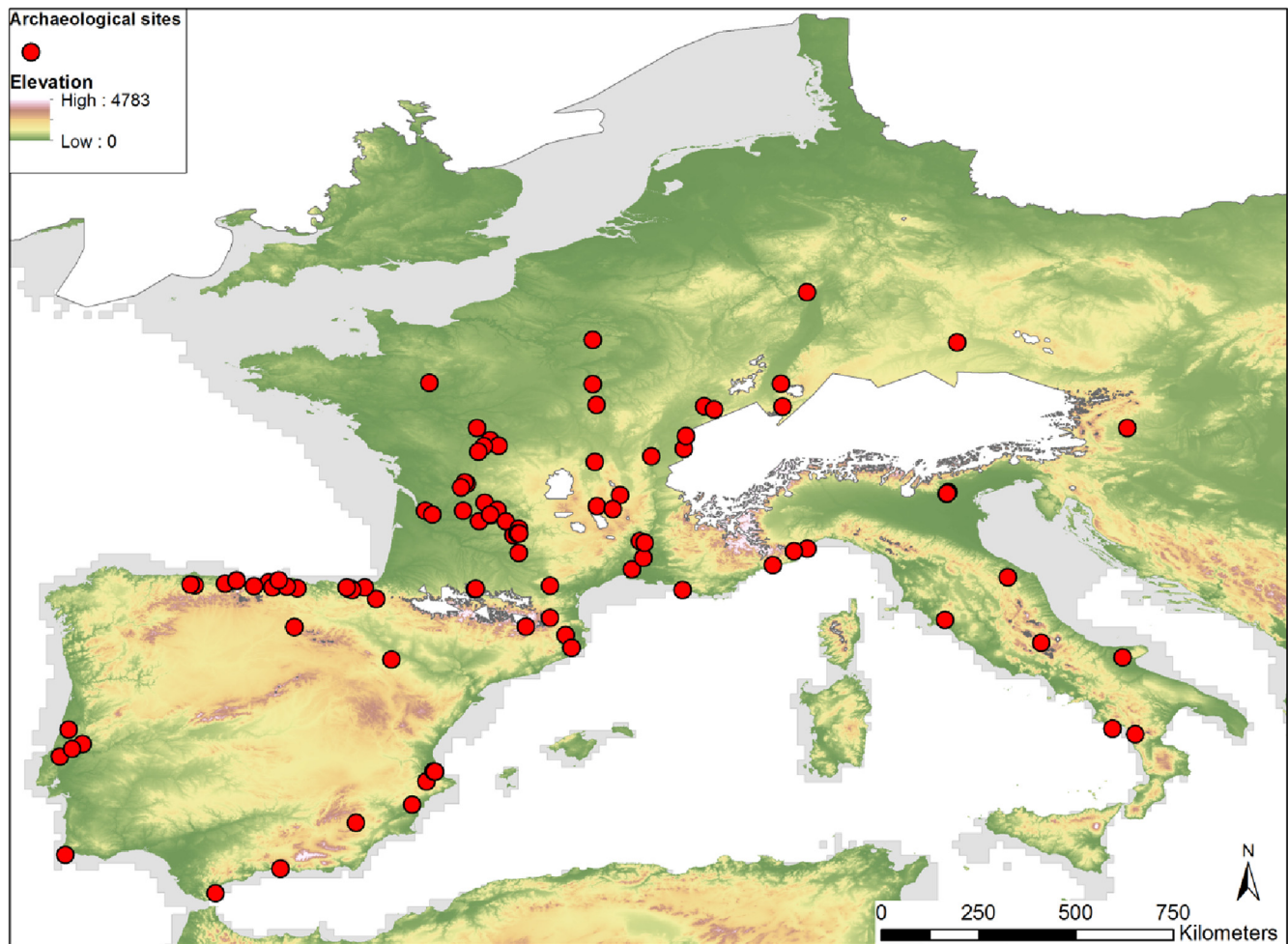


Fig. 1. The archaeological sample (presences). Western European archaeological sites dated to the LGM (red circles). The grey mask in the background represents the Pleistocene coastline at 21 ka; the white mask in the foreground represents the ice sheets, as per Ehlers et al. (2011). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and maxima are included since asymmetries exist in the distribution trends, e.g. between temperature minima and maxima (Beniston and Stephenson, 2004). Annual and seasonal average monthly cloud cover, interpolated from the AOGCM output on the high resolution grid, was also included.

We also calculated the dispersion of temperature and precipitation values from the 50-year run of simulated climate variables. For temperature we calculated the standard deviation (SD) of monthly temperature averages and for precipitation, the coefficient of variation (CV), i.e., the ratio of the standard deviation to the mean (Giorgi et al., 2004), averaged over monthly, seasonal and annual timespans. Next, we calculated the Standardized Precipitation Index (SPI) (Guttman, 1999; Hayes, 2000; McKee et al., 1993) using a subroutine of the “SPEI” R package (Vicente-Serrano et al., 2010) and a 12-month interval to standardize the values. The SPI calculations were used to establish the number of months when monthly average precipitation values fell above (dry) or below (wet) the thresholds defined by McKee (Vicente-Serrano et al., 2010) for “average”, “extreme” and “severe” conditions. The SPI is a drought index that is particularly suitable for our purposes because it only requires monthly precipitation values, while producing comparable (or identical) results to other, more complex indices (Chen et al., 2012: Table 1; Guttman, 1998). Finally, we calculated the Standardized Temperature Index (STI) using a purpose-built package for R based on the same principles as the SPI

(Fasel, 2014. STI: Calculation of the Standardized Temperature Index. R Package version 0.1. <https://cran.r-project.org/web/packages/STI/>). The resulting STI values were classified using the same scoring system as the SPI. The full list of candidate predictors ($N = 52$) is listed in Table 1.

2.4. Model fitting

We built a predictive model of the probability of encountering archaeological sites in a given location using Random Forests (RF). Random Forests (RF) is a nonparametric tool for variable selection and regression initially proposed by Breiman (2001). RF is an increasingly popular alternative to linear regression models because of its ability to deal with large numbers of predictors, relatively few observations and non-linear relationships (Genuer et al., 2010; Grömping, 2009). RF also provides useful measures of model performance and variable importance (VI) (Díaz-Urriarte and Alvarez de Andrés, 2006). Random forests are built by creating a large number of classification trees where each tree is based on a random subset of the observations, and each split within each tree is based on a random subset of m -try candidate variables (where m -try is the number of randomly selected predictors used at each node) (Grömping, 2009:311). The final model combines the results obtained for each tree within the forest.

First, we loaded the georeferenced archaeological sites (see above) into a point feature class. Rasters were produced for the 52 candidate predictors (Table 1) and resampled to a 1 km resolution using the Nearest Neighbour function in ArcGIS 10.3. The resulting shapefile was clipped to exclude the spatial extent of the ice sheets at 21 ka B.P. (Ehlers et al., 2011) from the analysis and all points within 1 km of a known archaeological site were designated as presences; all remaining points were designated as potential pseudo-absences. Predictor variables were then filtered in order to minimize multicollinearity using the findCorrelation function in the “caret” package in “R” (Kuhn et al., 2016). Using this function, we eliminated highly correlated predictors (i.e., those with $r \geq 0.8$). When two variables were found to be correlated at $r \geq 0.8$ the variable with the largest mean absolute correlation with all other

predictors was removed.

The resulting filtered list of variables ($N = 29$) was then used in a variable selection process using the RF function contained in the “caret” package. We used default values for the n -tree (the number of trees to grow; $N = 500$) and m -try ($\sqrt{p}$) parameters as recommended by Diaz-Uriarte & Alvarez de Andrés (Díaz-Uriarte and Alvarez de Andrés, 2006). In classification techniques such as RF the best results are often obtained by averaging several iterations (runs) and using fewer pseudo-absences than other regression techniques (Barbet-Massin et al., 2012). Therefore, we ran RF 100 times per run with 100 randomly selected pseudo-absences and 10-fold cross-validation. Model performance was assessed using the Receiver Operating Curve (ROC) during the model-tuning process. The “out of the bag” (OOB) error rate was calculated for all of the

Table 1

List of candidate predictors ($N = 52$) including the type of predictor (topographical, climatological, climate variability), the data source and a brief description.

Variable name	Type	Derivation	Description
altitude	Topographic	DEM	elevation (m asl)
cold_extrem	Climate variability	STI index	Temperature extreme (extremely cold)
cold_very	Climate variability	STI index	Temperature extreme (very cold)
dry_extrem	Climate variability	STI index	Precipitation extreme (extremely dry)
dry_very	Climate variability	STI index	Precipitation extreme (very dry)
hot_extrem	Climate variability	STI index	Temperature extreme (extremely hot)
hot_very	Climate variability	STI index	Temperature extreme (very hot)
p_avg_aut	Climatology	Climate simulation	Seasonal precipitation average, autumn
p_avg_spr	Climatology	Climate simulation	Seasonal precipitation average, spring
p_avg_sum	Climatology	Climate simulation	Seasonal precipitation average, summer
p_avg_win	Climatology	Climate simulation	Seasonal precipitation average, winter
p_avg_y	Climatology	Climate simulation	Annual precipitation average
p_max_aut	Climatology	Climate simulation	Seasonal precipitation maximum, autumn
p_max_spr	Climatology	Climate simulation	Seasonal precipitation maximum, spring
p_max_sum	Climatology	Climate simulation	Seasonal precipitation maximum, summer
p_max_win	Climatology	Climate simulation	Seasonal precipitation maximum, winter
p_max_y	Climatology	Climate simulation	Annual precipitation maximum
p_min_aut	Climatology	Climate simulation	Seasonal precipitation minimum, autumn
p_min_spr	Climatology	Climate simulation	Seasonal precipitation minimum, spring
p_min_sum	Climatology	Climate simulation	Seasonal precipitation minimum, summer
p_min_win	Climatology	Climate simulation	Seasonal precipitation minimum, winter
p_min_y	Climatology	Climate simulation	Annual precipitation minimum
p_var_aut	Climate variability	Climate simulation	Seasonal precipitation, coeff. var., autumn
p_var_spr	Climate variability	Climate simulation	Seasonal precipitation, coeff. var., spring
p_var_sum	Climate variability	Climate simulation	Seasonal precipitation, coeff. var., summer
p_var_win	Climate variability	Climate simulation	Seasonal precipitation, coeff. var., winter
p_var_y	Climate variability	Climate simulation	Seasonal precipitation, coeff. var., yearly
slope	Topographic	DEM	Slope (reclassified)
SPL_normal	Climate variability	Climate simulation	N months within normal (predicted) range
STI_norm	Climate variability	Climate simulation	N months within normal (predicted) range
sun_avg_aut	Climatology	Climate simulation	Seasonal average % cloudcover, autumn
sun_avg_spr	Climatology	Climate simulation	Seasonal average % cloudcover, spring
sun_avg_sum	Climatology	Climate simulation	Seasonal average % cloudcover, summer
sun_avg_win	Climatology	Climate simulation	Seasonal average % cloudcover, winter
sun_avg_y	Climatology	Climate simulation	Yearly average % cloudcover
t_avg_aut	Climatology	Climate simulation	Seasonal temperature average, autumn
t_avg_spr	Climatology	Climate simulation	Seasonal temperature average, spring
t_avg_sum	Climatology	Climate simulation	Seasonal temperature average, summer
t_avg_win	Climatology	Climate simulation	Seasonal temperature average, winter
t_avg_y	Climatology	Climate simulation	Yearly temperature average
t_max_aut	Climatology	Climate simulation	Seasonal temperature maximum, autumn
t_max_spr	Climatology	Climate simulation	Seasonal temperature maximum, spring
t_max_sum	Climatology	Climate simulation	Seasonal temperature maximum, summer
t_max_win	Climatology	Climate simulation	Seasonal temperature maximum, winter
t_min_aut	Climatology	Climate simulation	Seasonal temperature minimum, autumn
t_min_spr	Climatology	Climate simulation	Seasonal temperature minimum, spring
t_min_sum	Climatology	Climate simulation	Seasonal temperature minimum, summer
t_min_win	Climatology	Climate simulation	Seasonal temperature minimum, winter
t_sd_aut	Climate variability	Climate simulation	Standard deviation, seasonal temperature, autumn
t_sd_spr	Climate variability	Climate simulation	Standard deviation, seasonal temperature, spring
t_sd_sum	Climate variability	Climate simulation	Standard deviation, seasonal temperature, summer
t_sd_win	Climate variability	Climate simulation	Standard deviation, seasonal temperature, winter
t_sd_y	Climate variability	Climate simulation	Standard deviation, temperature, yearly
wet_extrem	Climate variability	Climate simulation	Precipitation extreme (extremely wet)
wet_very	Climate variability	Climate simulation	Precipitation extreme (very wet)

models in each run as a measure of model performance.

The variable importance (VI) index quantifies the relative importance of the predictors using the permutation accuracy importance measure (Strobl et al., 2007). Variables with high VI values force splits at higher nodes in the classification trees. Although not directly analogous to a standardized regression coefficient (the summary statistic used in logistic regression models) VI provides an indication of which variables are the most important in the final model. Average model performance and VI (over 100 iterations) were calculated for each run. Mean model predictions for each cell in the spatial extent were used to produce the final predictive model of archaeological site occurrence for each run.

The variable selection process consists of a nested series of RF runs that gradually winnows down the number of candidate predictors. The goal is to identify the most parsimonious list of predictors without compromising model performance. Candidate variables are ranked by importance (VI) at the end of each RF run

and the bottom 20% is removed from consideration in the succeeding run (Díaz-Urriarte and Alvarez de Andrés, 2006; Genuer et al., 2010). Model performance at the end of each run is evaluated using OOB. The model with the fewest predictors within u standard errors of the minimum error rate (OOB) for all of the RF runs is selected. When $u = 0$ this procedure selects the model with the best performance; when $u = 1$ it selects the model with the smallest number of predictors within one standard error of the best performance.

3. Results

The RF solutions resulting from each step of the variable selection process are illustrated in Fig. 2. Model performance (OOB) is illustrated in Tables 2 and 3 provides the Variable Importance (VI) index (a measure of relative importance) for the individual predictors for each iteration.

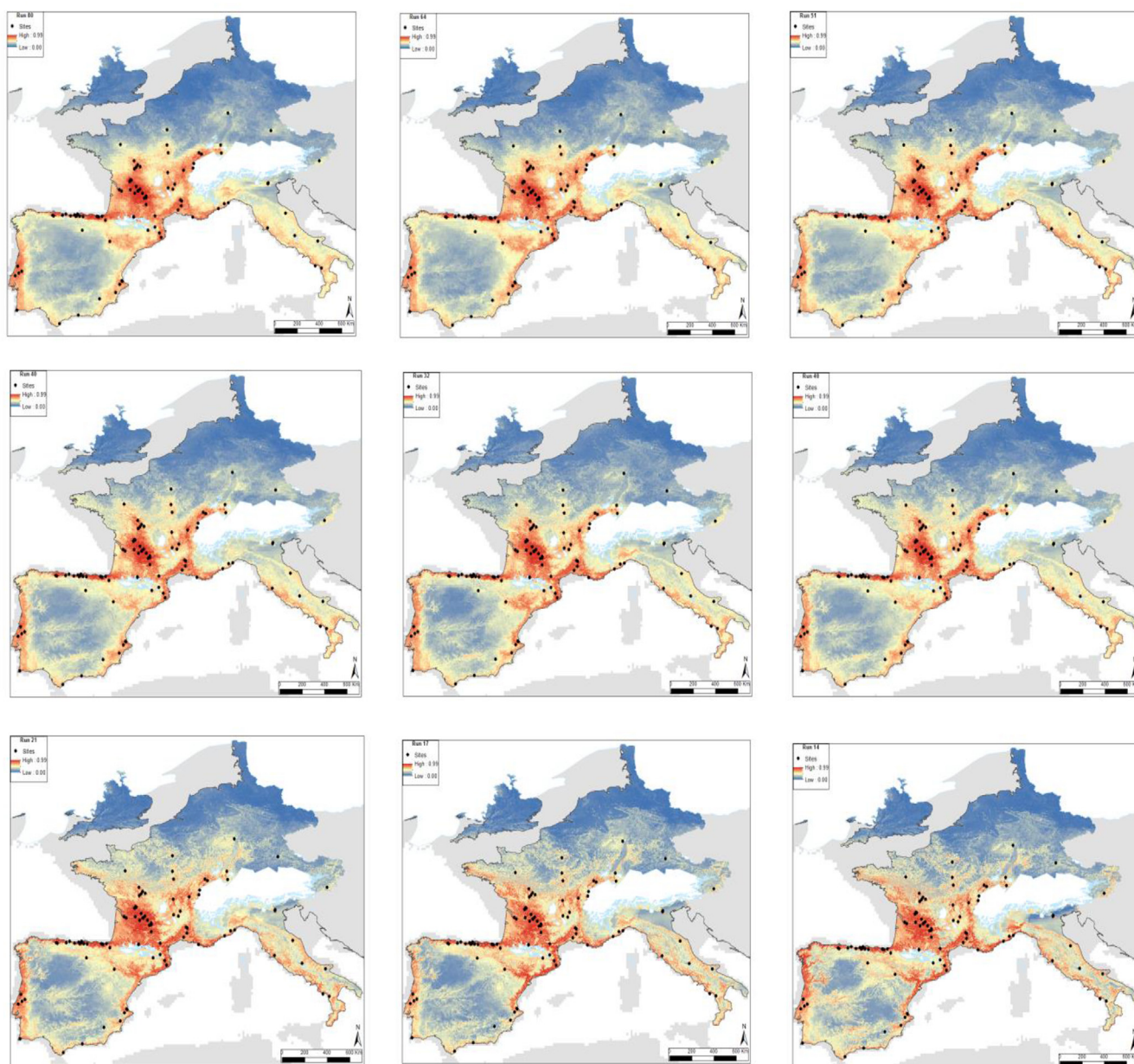


Fig. 2. Predictive models produced at each step of the variable selection process, from run 80 to run 14.

Model performance (OOB) at each step of the model selection process (s.e.: 0.029).

Run	N variables	OOB
Run 100	26	0.269
Run 80	21	0.267
Run 64	17	0.272
Run 51	14	0.267
Run 40	12	0.269
Run 32	9	0.270
Run 26	7	0.266
Run 21	6	0.276
Run 17	5	0.276
Run 14	4	0.293
Run 11	3	0.329
Run 9	2	0.359

The predictors included in model 26 are: altitude, slope, p_var_spr (the coefficient of variability for Spring precipitation rates), t_max_aut (maximum monthly average temperatures for Autumn), STI_norm (variability in monthly temperatures calculated using the STI index), t_sd_win and t_sd_aut (standard deviation in average monthly temperatures in winter and in autumn).

The set of predictors selected in our habitat suitability model (model 26) includes topographical (altitude, slope) and climatological variables (`t_max_aut`) as well as indices of climate variability (`p_var_spr`, `STI_norm`, `t_sd_win` and `t_sd_aut`). These results are consistent with our initial findings based on a more geographically limited study (Burke et al., 2014). Broad latitudinal differences in the distribution of some of the predictors are apparent (Fig. 5), e.g., `t_max_aut` and `p_var_spr`, illustrating the complexity of a continental-scale dataset. The total set of candidate variables considered in this analysis contains multiple correlations (S2 Table 1) and interactions between variables are likely to occur (e.g., temperature and precipitation rates). Some generalisations about the distribution of archaeological sites against background values are possible, for example: sites are distributed at a mean elevation of 257 m *asl* (whereas mean elevation for absences is 411 m *asl*), on gently sloping ground ($<27^\circ$), in regions where

[illegible]

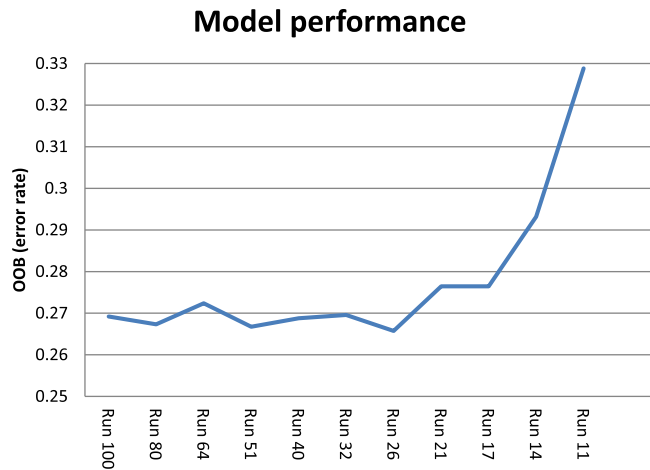


Fig. 3. Model performance (OOB) for each run in the variable selection process.

maximum autumn temperatures are above the continental mean (ranging between 8° and 17.7 °C) while variation in spring precipitation is slightly below the mean but well below maximum possible values for the background region. Fig. 6 illustrates density values for the standardized variables, comparing presences and absences.

The lowest habitat suitability scores associated with archaeological sites occur in Northern Europe where three sites ringing the Alps are associated with small, isolated clusters of relatively suitable habitat, including: Wiesbaden Ingstadt (54.4%), Mittlere Klaus (57.5%) and Grusgraben (56%) (Fig. 4). The paucity of evidence for human occupation in Northern Europe is not the result of a lack of research effort, particularly in Germany – a country with a long tradition of archaeological surveys. Instead, our model suggests that the distribution of suitable habitat was simply too sparse in this region to support a large population during the LGM. In Southwestern France, conversely, our model predicts that the Landes region would have been very suitable habitat (~60–70%). What appears to be a poor prediction given the lack of evidence for human occupation in the Landes during the LGM (Fig. 4) may be explained by a variable not included in our model, namely soil type. Recent research indicates that the region was covered with sand, with low-relief dune fields surrounded by loess accumulations creating a periglacial desert (Bertran et al., 2013). The predictive model illustrated above (Fig. 4) should be interpreted broadly, therefore, and adjusted for local factors.

Generally speaking, topographic variables (altitude, slope) are important predictors of site location, followed by relatively warm autumnal temperatures and variation in spring precipitation rates, which affect the length of the growing season and the availability of plant and animal resources (i.e., productivity). All things being equal, seasonal variation in temperature (t_{sd_win} and t_{sd_aut}) and climate variability (STI_norm) also influence habitat suitability. At a continental scale, therefore, inter-annual variation in climate conditions and temperature variability – variables we consider to represent ecological risk – may have played an important role in making some regions of Europe more attractive to groups of foragers than others.

Finally, our model suggests that suitable habitat is not uniformly distributed in Western Europe, nor is it limited to the southern Mediterranean peninsulas. Rather, the picture is one of a fragmented landscape, with relatively large extents of unsuitable habitat broken up by smaller, loosely connected clusters of suitable habitat. The implications of this spatial fragmentation in demographic terms are discussed below, but it also has potential

repercussions for cultural diversity. Recent modelling experiments, for example, have shown that partially connected groups produce more diverse solutions to the problems they face (Derech and Boyd, 2016). This is something we hope to test in the very near future.

4.1. Model comparisons

Other researchers have modelled the impact of environmental conditions on human population dynamics in Europe during the LGM (e.g., Banks et al., 2008; Tallavaara et al., 2015). Their methodology differs somewhat from ours in terms of the climate simulations used and the predictors included in the analyses. Tallavaara et al. use the Climber 2 climate simulation and climate predictors exclusively, while Banks et al. use a simulation run with IPSL-CM4 (an earlier version of the model used here) and a combination of climate predictors and topographical predictors (as we do) but segregate the presence data by archaeological culture.

Tallavaara et al. (2015) derive a model of habitat suitability for hunter-gatherers using ethnohistorical and ethnographic data and present-day climate conditions. This model is then deployed to predict the distribution and density of human populations for a series of time slices using simulated climate conditions. The resulting model of population distributions during the LGM in Europe (Tallavaara et al., 2015: Fig. 2) predicts a relatively large human population in the Iberian Peninsula concentrated in the south, along the Mediterranean coastline and in the interior (along the Guadiana River). Conversely, they do not predict human occupation north of the Dordogne. As the authors themselves point out, these predictions are not compatible with the archaeological record as we know it (see Fig. 1, above). In contrast, our models suggest that the interior of the Iberian Peninsula was relatively unsuitable habitat, while Cantabria and both the Atlantic and Mediterranean seaboard were eminently suitable, as were parts of France north of the Dordogne (e.g., the southern reaches of the Paris Basin). The differences between our model predictions and predictions made by Tallavaara et al. (2015) probably reflects limitations in the ethnographic record and the fact that the authors do not incorporate Inuit or Athapaskan groups in the modelling process. This could explain why low temperatures are a limiting factor in Tallavaara et al.'s model, whereas they are present but relatively more relaxed in our model. The observable differences between models could also stem from the use of a different climate simulation (Climber 2) to model conditions during the LGM. Finally, the absence of key predictors (climate variability, topographic variables) could also explain why Tallavaara et al.'s model predictions are not compatible with ours. Our results, in contrast, reflect the resilience of Upper Palaeolithic people and their ability to use technological innovation and social strategies to overcome physiological limitations and inhabit more northerly latitudes.

Banks et al. (2008) use archaeological site distributions as presence data and machine-learning techniques to reconstruct the “ecocultural niche” of two human populations associated with distinct archaeological cultures (the Solutrean and the Epi-gravettian) during the LGM. Instead of Random Forests, however, the authors use a general algorithm for rule prediction (GARP). The resulting models (Banks et al., 2008: Fig. 4) compare well with ours, but there are differences in the model predictions, most notably in the centre of the Iberian Peninsula. These differences could be due to cultural preferences (as the authors suggest), or they could reflect differences in the candidate predictors selected for the analysis and finally, the decision to model Solutrean and Epi-gravettian populations separately. There are also basic differences in the climate simulation used as well as the downscaling methodology (Banks et al. use the LMDZ with a “zoom” over Europe) which results in a slightly coarser spatial resolution than the data

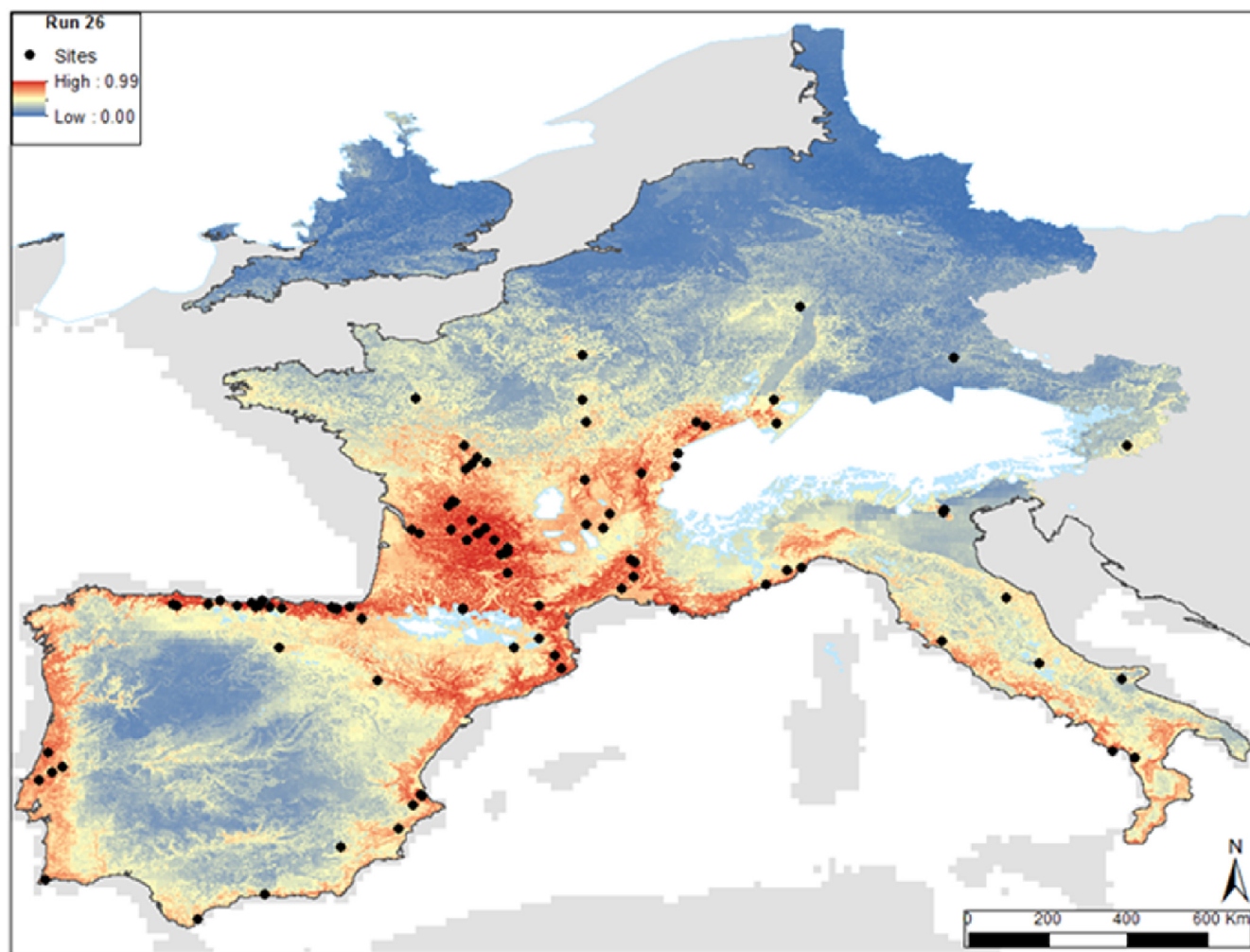


Fig. 4. The final habitat suitability model (model 26). Archaeological sites used as presences in the analysis are represented by black dots against a background of habitat suitability.

produced for this research. Finally, we included several candidate predictors that quantify climate variability in our initial set of variables and treat all contemporaneous (LGM) occupations as a single set of presence data. The goal, in our case, is to predict the distribution of human populations rather than describe the environmental conditions under which individual archaeological entities existed.

5. Conclusions

The ethnographic record indicates that modern hunter-gatherers are sensitive to ecological risk, deploying a variety of strategies to counter its effects. These strategies include the adoption of different patterns of mobility, such as logistical mobility (Riel-Salvatore and Barton, 2004) and the use of social mechanisms such as information-sharing and reciprocity. These measures have limitations, however, depending on the temporal and spatial scale of the risk. In addition to facing rigorous climate conditions and ecological risk during the LGM, a decline in human population size (Bocquet-Appel et al., 2005; Davies and Gollop, 2003; Morin, 2008; Tallavaara et al., 2015) may have hampered the ability of foraging groups to maintain effective social “safety nets” to act as hedges against unpredictable resource distribution. When all else fails human populations disperse, seeking more productive and less ecologically risky territories elsewhere.

Our results suggest that northern regions of Western Europe were indeed largely unsuitable for human occupation during the LGM as a result of the combined effect of ecological risk, climate conditions and topography. Suitable conditions for human habitation existed in Franco-Cantabria, along the Atlantic and Mediterranean seaboard of the Iberian Peninsula, in Liguria and along the Rhone River corridor. Small pockets of relatively suitable habitat existed in some more northerly regions, including the southern margin of the Paris Basin, the Rhone valley and parts of the Jura. These “micro-refugia” (or cryptic refugia) are connected (sometimes tenuously) to larger, more southerly clusters of suitable habitat via major river valleys. There is little reason to suggest that these regions supported a large population (in this we agree with Tallavaara et al., 2015), or even that they were occupied continuously. Nevertheless, there is growing archaeological evidence for the existence of “micro-refugia”, i.e., human presence in more northerly regions once thought to have been abandoned under full glacial conditions (Aubry et al., 2012; Bodu et al., 2007; Debout et al., 2012; Terberger and Street, 2003).

Other biogeographical evidence also supports the existence of cryptic refugia for a variety of plants and animals in more northerly regions of Western and Central Europe during the LGM (Bhagwat and Willis, 2008; Boston et al., 2014; Hofreiter and Stewart, 2009; Provan and Bennett, 2008; Sommer and Benecke, 2005; Willis et al., 2002). It has been suggested that the existence of “micro-

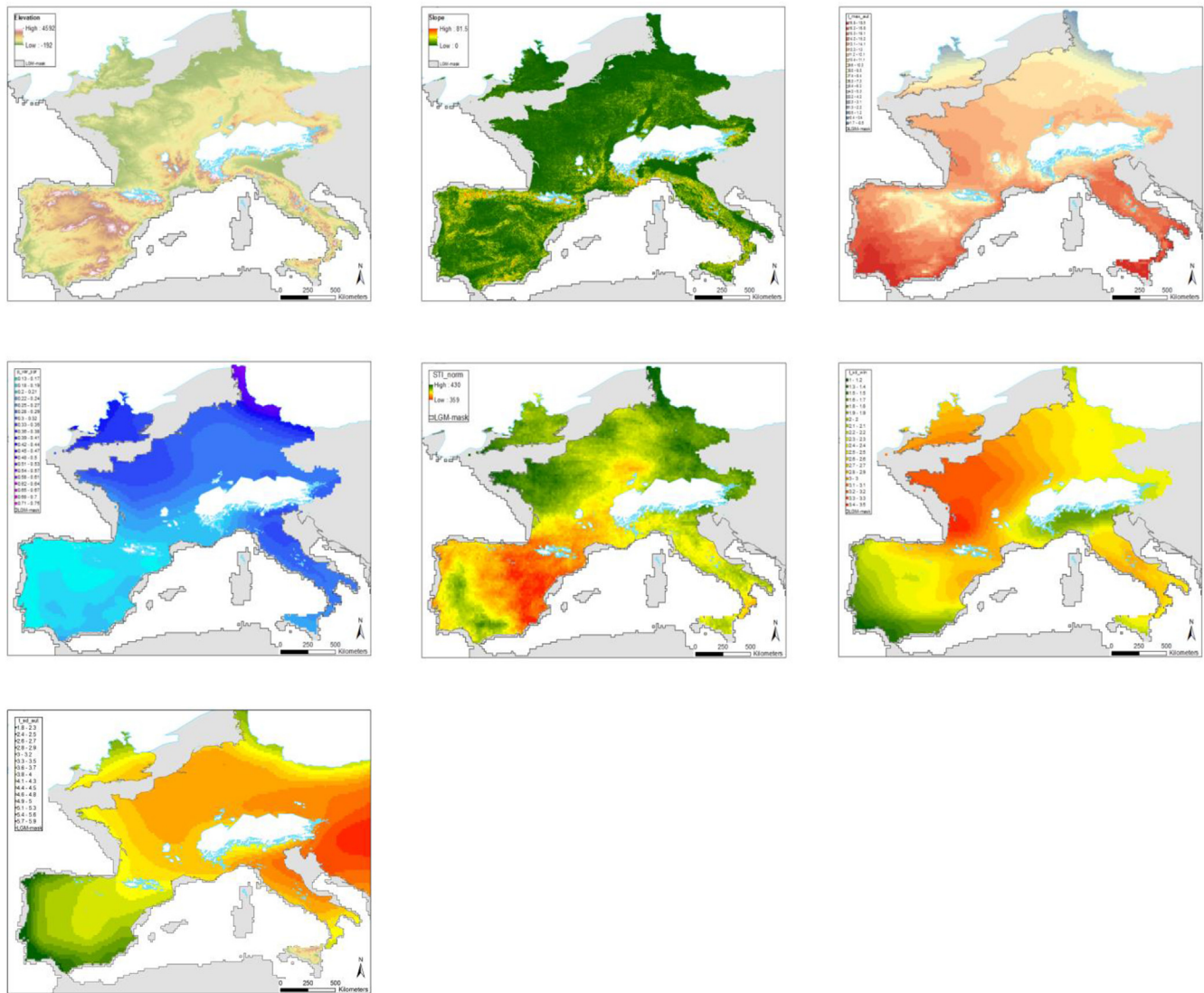


Fig. 5. Spatial distribution of the 7 predictors included in model 26, including: 1) elevation; 2) slope; 3) maximum temperatures in autumn ($t_{\max_aut}$); 4) variation in spring precipitation rates (p_var_spr); 5) temperature variability (STI_norm); 6) variation in winter temperature (t_sd_winter); 7) variation in autumn temperature (t_sd_aut).

refugia" north of Mediterranean Europe could explain the apparently rapid re-colonisation of Western and Central Europe at the end of the LGM (Cupillard and Welte, 2006; Demars, 2002; Gamble et al., 2006; Hublin, 2014; Rensink, 2012). Larger clusters of suitable habitat to the south may have acted as "sources" for periodic population expansions into adjacent, less suitable regions, such as the proposed "micro-refugia". These smaller regions, in turn, may have acted as population "sinks", periodically emptying as climate conditions fluctuated. Source populations may have been seasonally, or perhaps only sporadically present in these "sinks". As the climate began to ameliorate at the end of the LGM, the "sinks" would have been the first regions to have been re-colonised, acting as conduits for further expansion into northern Europe. A similar model of "sinks" and "sources" is proposed for the Middle Pleistocene of Europe (Dennell et al., 2011) as an alternative to models such as "ebb and flow" (in which populations retreat to refugia when conditions worsen) or "niche tracking" (in which groups simply track appropriate environmental conditions) and fits the discontinuous pattern of Neanderthal occupation in Northern Europe (Hublin and Roebroeks, 2009).

In conclusion, this analysis demonstrates the importance of topography, climate conditions and climate variability at fine temporal and spatial scales as drivers of human spatial behaviour during the LGM. Our predictive models suggest that regions with suitable habitat, i.e., inherently less "risky" and relatively productive regions, are discontinuously distributed across Western Europe during the LGM when they tend to concentrate along coastal plains and in low-lying regions interconnected by river valleys. The spatial discontinuities between suitable "micro-habitats" could explain the observed "regionalisation" of Upper Paleolithic cultures during the LGM, since reduced mobility across unpredictable landscapes would have restricted the flow of people and thus ideas, objects and genes between regional populations. The impact of climate variability at fine spatial and temporal scales, therefore, extends beyond the spatial distribution of populations, influencing their cultural and biological evolution as well. Finally, the spatial models developed during the course of this research will be used to design and carry out archaeological surveys in the near future, which will allow us to test their predictive power, while contributing to improving survey design in

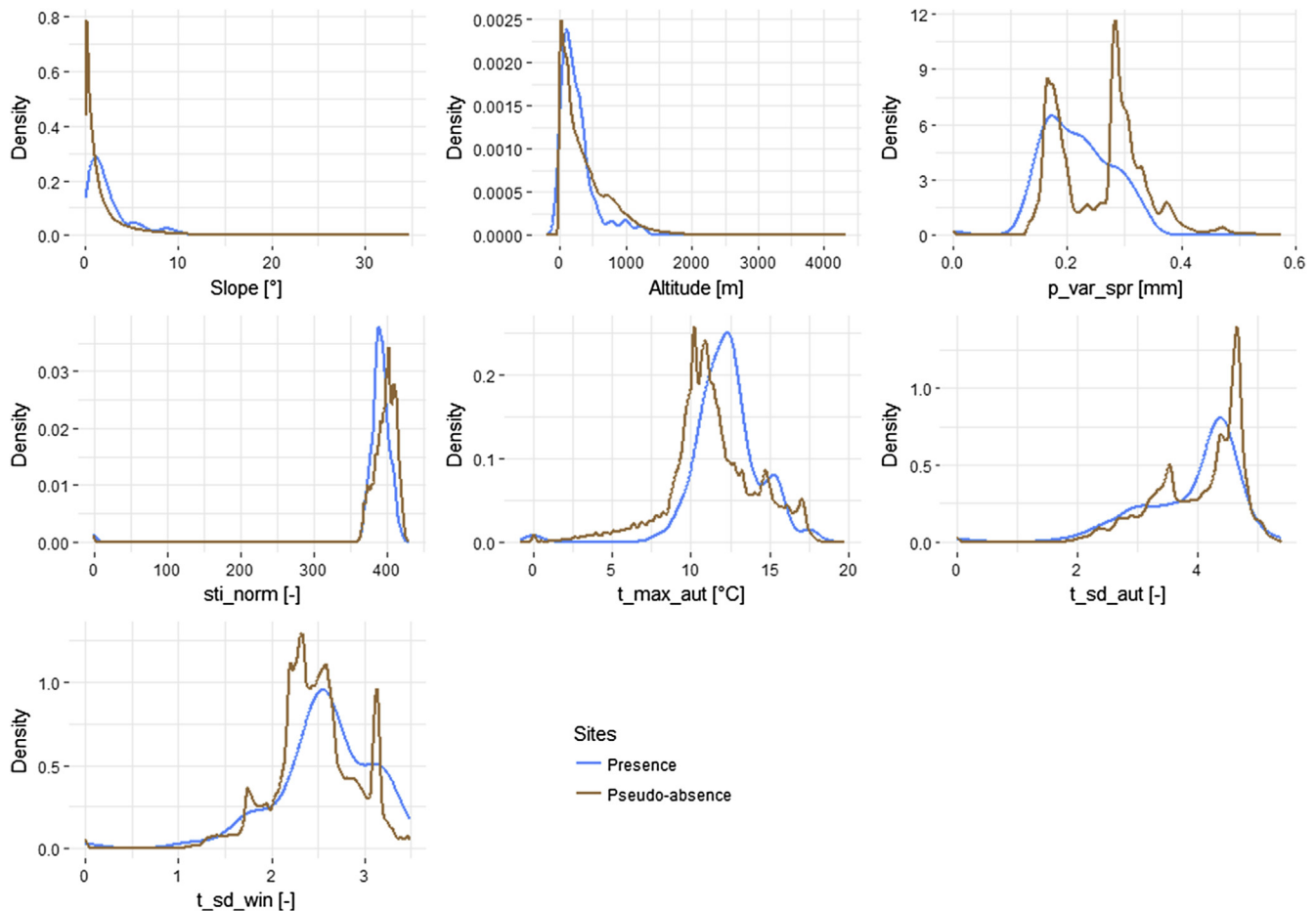


Fig. 6. Density plots (presences and pseudo-absences) for the 7 predictors included in the habitat suitability model (model 26).

regions where the archaeological record is relatively poorly documented.

Acknowledgements

This research was supported by the Fédération Québécoise de Recherche en Sciences Sociales - Programme de soutien aux équipes (demande no. 179537). The authors would like to acknowledge the valuable assistance of the Direction Régionale de l'Archéologie et de la Culture (DRAC): Hélène Breichner, Olivier Bigot, Pascale Barthes, Pascal Foucher, René Liabeuf, Yves Pautrat, Frédérique Juhaud, and all of the regional archaeologists who helped us refine the database, including Jean-Christophe Castel and Marie-Isabelle Cattin (Switzerland); Thierry Aubry, Pierre Chalard-Biberson, Christophe Cupillard, Cyril Montoya, Pierre Antoine and Olivier Moine (France); Martin Street, Andrew Kandel and Suzanne Munzel (Germany); Marguerita Mussi, Julien Riel-Salvatore, Marco Peresani and Matteo Romandini (Italy); Marcel Otte (Belgium) and Jonathan Haws (Portugal). William Davies kindly made the S2Ages database available to us and we gratefully acknowledge the contribution of the PACEA database (d'Errico et al., 2011). We apologize for any omissions we may have made inadvertently to this list of collaborators.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.quascirev.2017.04.001>.

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